

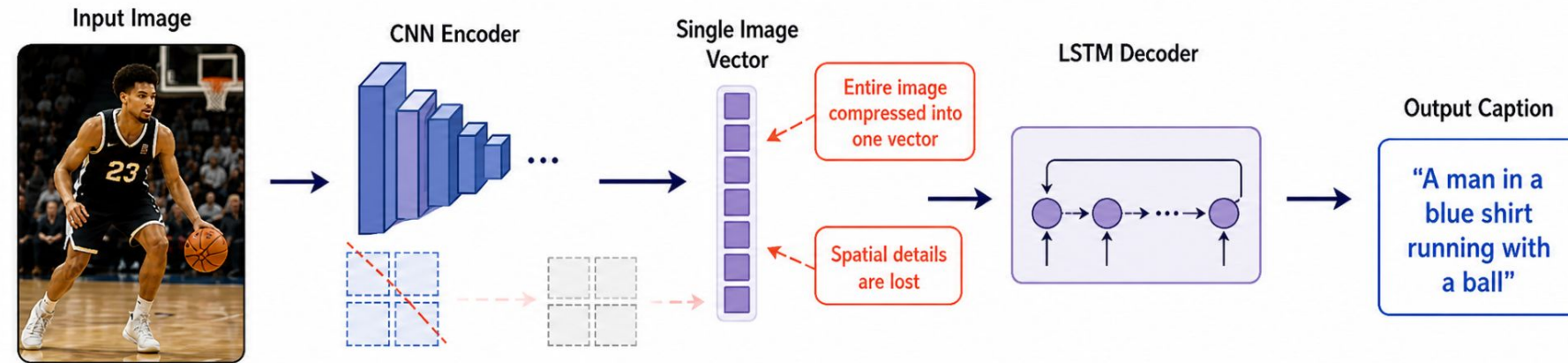
Show, Attend and Tell: Neural Image Caption Generation with Visual Attention

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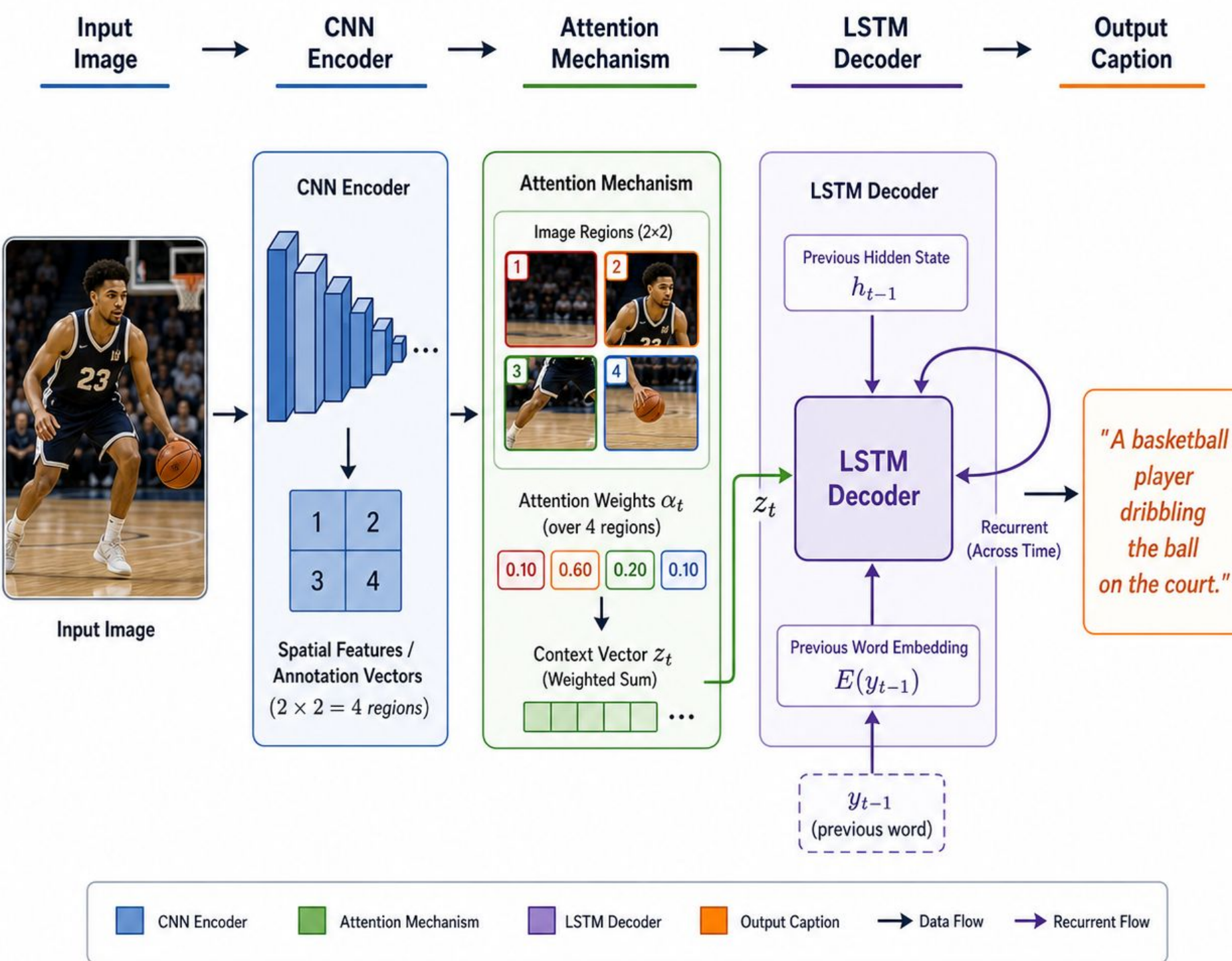
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Introduction

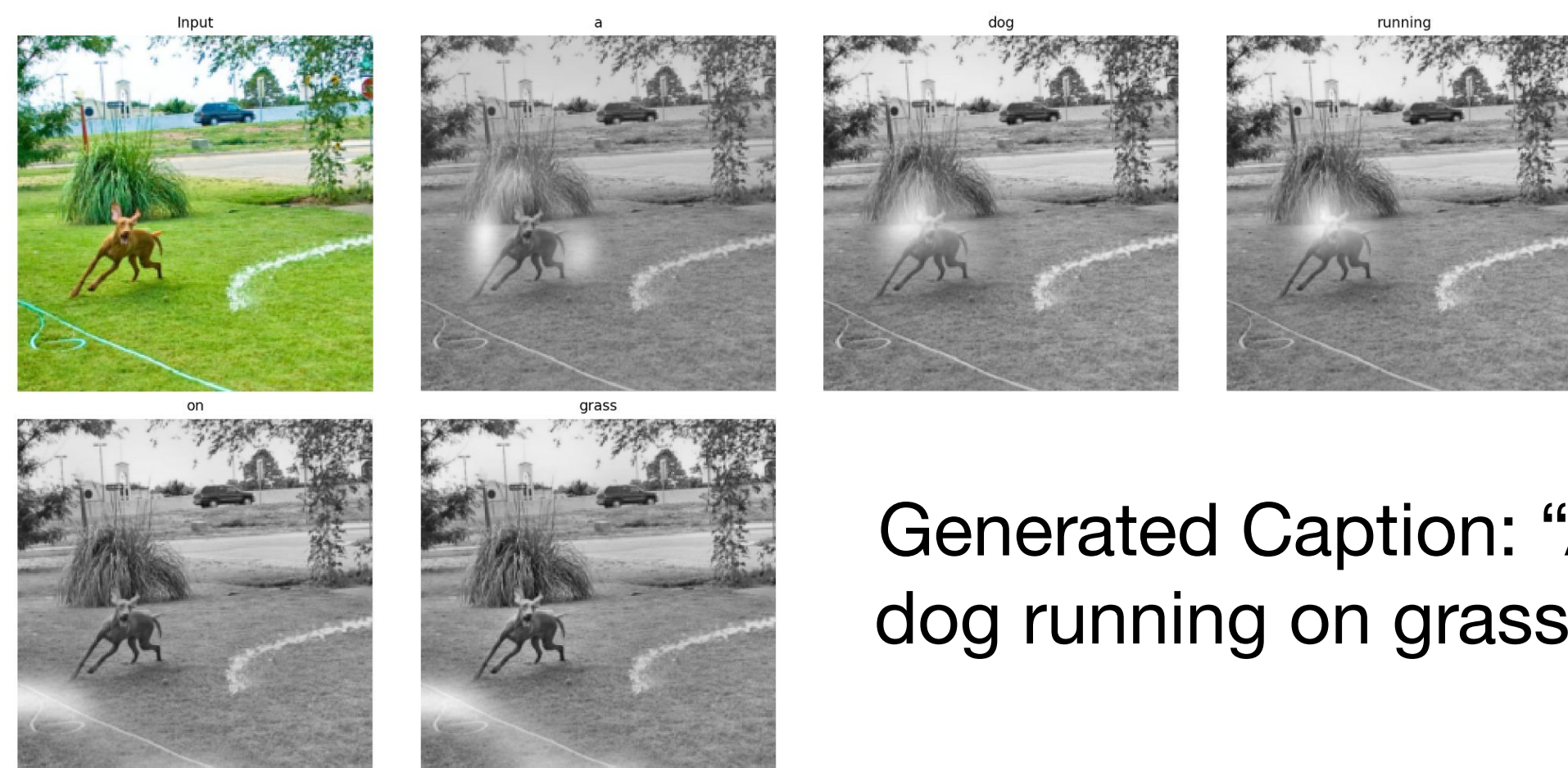
- Traditional encoder-decoder models compress an entire image into a single vector, losing the spatial information needed for accurate captioning.
- Show, Attend and Tell solves this by preserving spatial feature maps and using attention to focus on relevant image regions.



Architecture



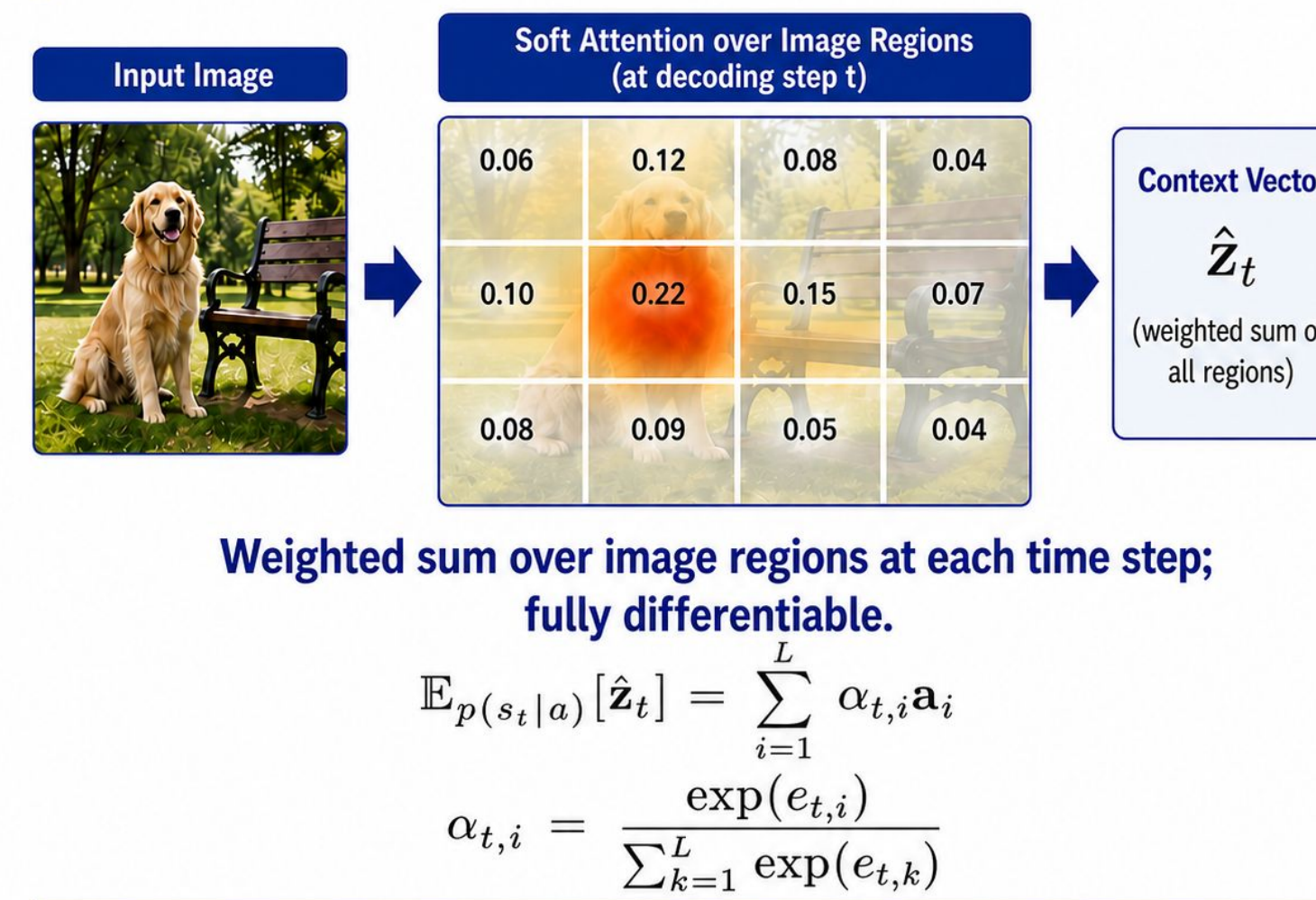
Visualization



Generated Caption: "A dog running on grass."

Methodology

- Soft Attention
 - Includes weighted average of **all image regions**
- Flickr8K Dataset
- BLEU Evaluation
 - Primary metric used to **quantitatively validate caption quality**
- VGG-19 & VGG-16 Pre-Trained Encoder CNN
 - Creates spatial feature maps
- LSTM-Decoder

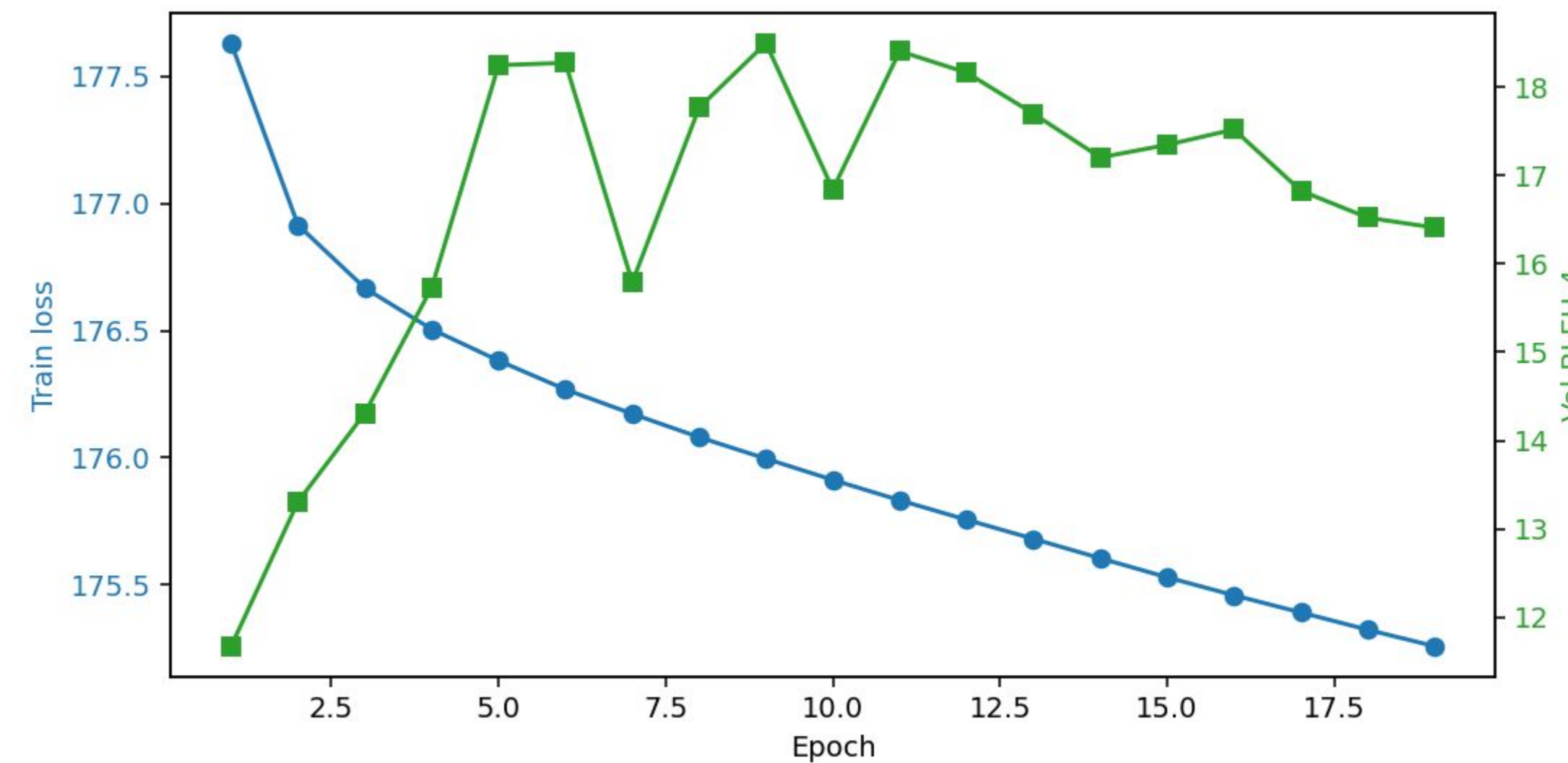


Doubly Stochastic Attention

$$L_d = -\log(P(y|x)) + \lambda \sum_{i=1}^L \left(1 - \sum_{t=1}^C \alpha_{t,i}\right)^2$$

This regularization encourages the model to attend to every image region over the full caption generation process.

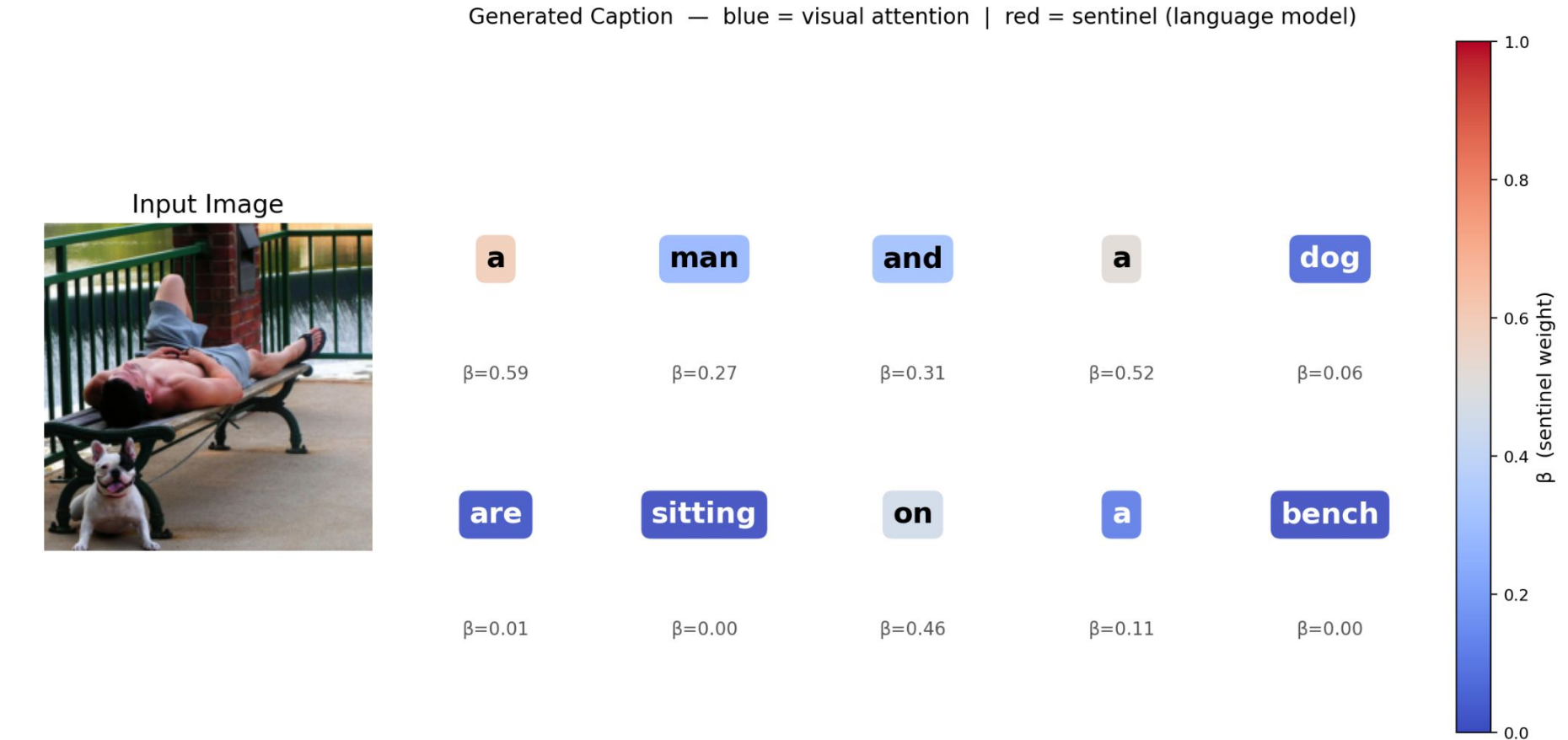
Results



Model / Decoding Setting	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Soft Attention	67.00	44.80	29.90	19.50
Hard Attention (paper)	67.00	45.70	31.40	21.30
Exact Replication, Beam=1, no length norm.	60.64	41.93	27.89	18.36
Optimized Replication, Beam=1, no length norm.	60.95	42.95	29.13	19.53
Experiment, Beam=3, no length norm.	64.72	46.52	32.56	22.27
Experiment, Beam=3, length norm.	61.20	43.76	30.45	20.97
Experiment, Beam=5, no length norm.	66.97	48.62	34.06	23.48
Experiment, Beam=5, length norm.	61.32	44.00	30.70	21.22

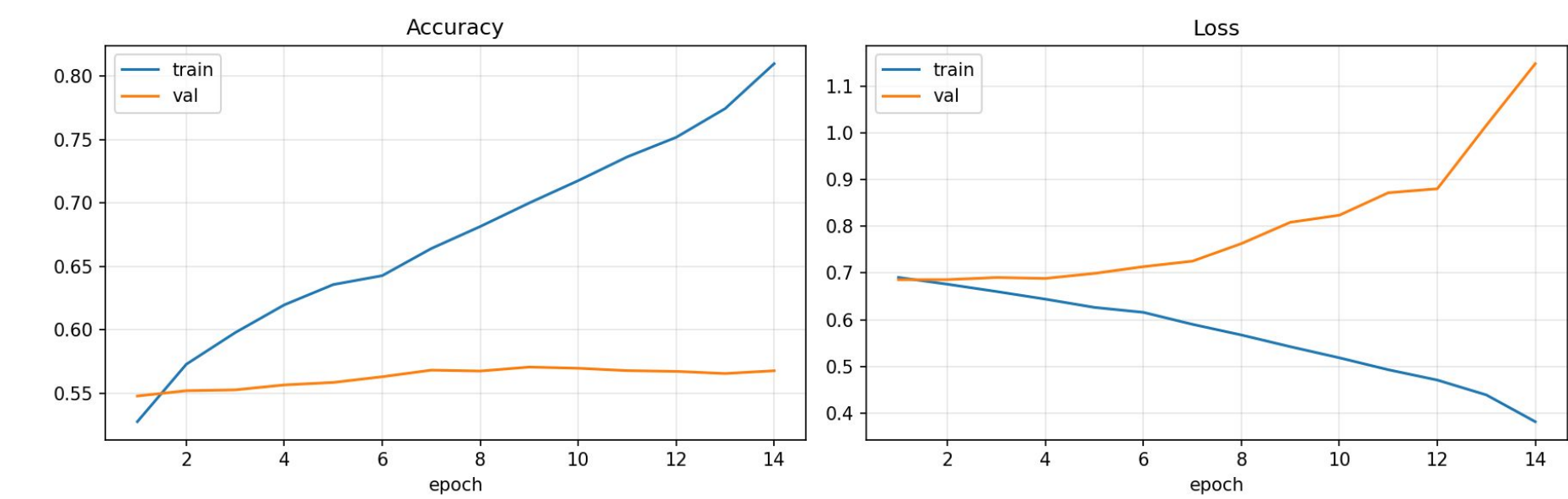
Exact Replication follows the original paper setup, while Optimized Replication and experiments use VGG16 fine-tuning, Adam, a learning rate scheduler, and gradient clipping for improved training stability.

Adaptive Attention



Model / Decoding Setting	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Soft Attention (paper)	67.00	44.80	29.90	19.50
Hard Attention (paper)	67.00	45.70	31.40	21.30
Our Adaptive Attention $\lambda=1.0$	62.40	43.60	29.10	18.80
Our Adaptive Attention $\lambda=0.01$	62.20	43.40	29.40	19.80

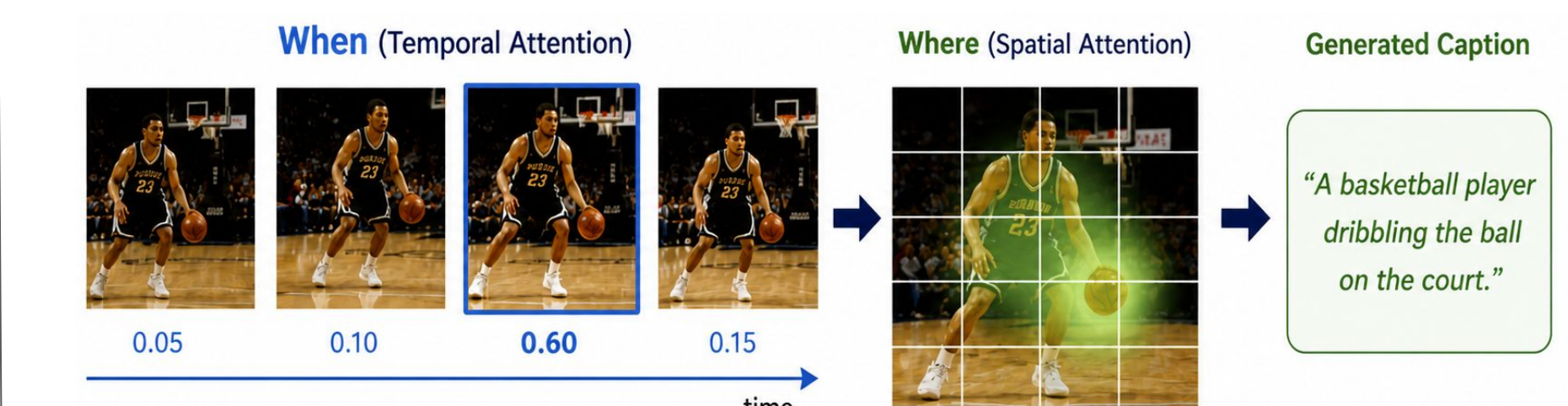
Visual Question Answering



The diverging validation loss suggests that collapsing the recurrent attention decoder into one fixed step limits generalization.

Conclusion & Future Directions

- Attention enhances interpretability and quality of image captions.
- Beam search, length normalization, and optimizer tuning improved BLEU over the exact replication.
- Video Captioning in the Future?



References

Xu, K., Ba, J. L., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R. S., & Bengio, Y. (2015). Show, attend and tell: Neural image caption generation with visual attention. Proceedings of the 32nd International Conference on Machine Learning.
Lu, J., Xiong, C., Parikh, D., & Socher, R. (2017). Knowing when to look: Adaptive attention via a visual sentinel for image captioning. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 375–383.